實習生月報

PDO 資料路徑重建

CoE / SDO 交易重建

PDO Mapping / Assignment 關聯整理

SyncManager / FMMU 資料路徑分析

報告期間

2026.08

MONTHLY REPORT

范舜傑 | MDSBU 通訊工程課 | 08/25/2026

DELTA LOGO

本月工作大綱

7/30 – 8/5

理解封包與 EtherCAT 100%

- 使用 Wireshark 觀察從上電抓的原始 EtherCAT 封包
- 整理 Broadcast、AP、FP 等指令對 Slave 的操作方式
- 整理 Master 對 Slave ESC 暫存器的讀寫流程
- 理解 CoE、SDO、PDO 與 Object Dictionary 的關係
- 建立本月主線：CoE / SDO → PDO Mapping / Assignment → SyncManager → FMMU

8/6 – 8/12

建立 Slave 拓樸關係 80%

- 理解 Master 如何透過 Auto Increment / Configured Address 指定 Slave
- 理解 Configured Station Address 與 Configured Station Alias 的差異
- 設計 AP Datagram 配對、Slave Discovery 模型與分析流程

設計 Analyzer 驗證 Agent 20%

- 建立 Stage 3 Result Check
- 由 Capture evidence 驗證 Slave Count、Topology、Address 與 Identity

8/13 – 8/19

Mailbox / CoE / SDO 70%

- 在 Datagram Record 中加入 CoE / SDO decoded fields
- 重建 SDO Initiate Download Request / Response Transaction
- 依 SDO Index / SubIndex 辨識 PDO Mapping / Assignment 操作
- 重建 RxPDO / TxPDO Mapping Configuration
- 重建 0x1C12 / 0x1C13 PDO Assignment Configuration

完善 Agent 驗證流程 30%

- 整理 TShark EtherCAT / CoE / SDO 查詢與 JSON 結構規格
- 將常用 Capture 查詢封裝成 Python tools，提供 LLM 多種驗證方式

8/20 – 8/21

SyncManager / FMMU 分析 100%

- 在 Datagram Record 中加入 SyncManager / FMMU decoded fields
- 重建 SyncManager Configuration
- 重建 FMMU Configuration
- 驗證 PDO Assignment 與 SM2 / SM3 的設定關係
- 驗證 FMMU Physical Address 與 SyncManager RAM 區段的對應

SECTION / 01

01

上電流程與 ESC Register 解析

追蹤啟動封包，建立 ESC 暫存器存取與解析脈絡

上電封包操作流程

01 Broadcast Initialization
BRD / BWR / BRW
All Slaves

02 Auto Increment Addressing
APRD / APWR / APRW
Topology Position

03 Configured Station Address
APWR / ADO 0x0010
Fixed Address

04 Fixed Position ESC Access
FPRD / FPWR / FPRW
Configured Address + ADO

Master

Slave 1
Outgoing ADP: 0x0000
Signed: 0
Topology Position: 1

Slave 2
Outgoing ADP: 0xFFFF
Signed: -1
Topology Position: 2

Slave 3
Outgoing ADP: 0xFFFE
Signed: -2
Topology Position: 3

Outgoing ADP	Signed Value	命中的拓樸位置
0x0000	0	Slave 1
0xFFFF	-1	Slave 2
0xFFFE	-2	Slave 3

Master 以 outgoing ADP = 0、-1、-2... 逐站指定 Slave。

ADO 是目標 ESC Register Offset，命中的 Slave 會執行該 Datagram。

SII EEPROM 讀取流程

01 **檢查 EEPROM 狀態**

- 讀取 ESC Register `0x0502~0x0503`
- 確認 Busy 清除且沒有 Error

02 **指定 SII Word Address**

- 寫入 ESC Register `0x0504~0x0507`
- `0x0008` = Vendor ID，`0x000A` = Product Code 起始 Word

03 **送出 EEPROM Read Command**

- 在 `0x0502~0x0503` 的 Control / Status 欄位送出 Read Command

04 **讀取 EEPROM Data**

- 以 APRD 讀取 `ADO = 0x0508`
- 依前面指定的 SII Word Address 解讀資料內容

Address Space	Address	Meaning
ESC Register	0x0502~0x0503	EEPROM Control / Status
ESC Register	0x0504~0x0507	EEPROM Address
ESC Register	0x0508~0x050F	EEPROM Data
SII EEPROM Word	0x0008~0x0009	Vendor ID
SII EEPROM Word	0x000A~0x000B	Product Code

實際 Capture

實際 Capture 中，Master 會以一個 `APWR` Datagram，從 `ADO = 0x0502` 開始一次寫入 `6 Bytes`。前 2 Bytes 對應 `0x0502~0x0503` 的 EEPROM Control / Status，後 4 Bytes 對應 `0x0504~0x0507` 的 EEPROM Address，因此一次 `APWR` 即可同時送出 Read Command 與欲讀取的 SII Word Address。

EEPROM Read 完成後，Master 再以 `APRD` 讀取 `ADO = 0x0508` 的 EEPROM Data。實際取得的資料內容，則由前面寫入的 SII Word Address 決定，例如 `0x0008` 對應 Vendor ID、`0x000A` 對應 Product Code。

Slave Discovery 重建成果

Slave Discovery

- AP Datagram
- Topology Position
- Configured Address
- Slave Identity

SECTION / 02

02 CoE / SDO 與 PDO Mapping 重建

從 CoE / SDO Transaction 重建 PDO Mapping 與 Assignment，建立 Process Data 的資料定義。

CoE / SDO 知識整理

01 **Mailbox / CoE / SDO**
Protocol Layers

02 **Object Dictionary**
Index / SubIndex

03 **SDO Download / Upload**
Client / Server

04 **SDO → PDO Configuration**
PDO Mapping

CoE / SDO 知識主線

- EtherCAT Datagram
- Mailbox
- CoE
- SDO
- Object Dictionary
- PDO Configuration

CoE / SDO 的作用是透過 EtherCAT Mailbox 存取 CANopen Object Dictionary，並可進一步用來設定 PDO Mapping / Assignment。

SDO Transaction / PDO Mapping 重建成果

EtherCATAnalyzer.TestConsole.exe analyze　　Stage 4 / 105 Transactions

```
[Stage 4] SDO Transaction Analysis
Elapsed: 6.226 s
Transaction Count: 105
Configured Slave Address: 0x0001, Object: 0x1C12:00, Data: 0x00, Request Frame: 41394, Response Frame: 41403, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1C13:00, Data: 0x00, Request Frame: 41408, Response Frame: 41417, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:00, Data: 0x00, Request Frame: 41422, Response Frame: 41455, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:01, Data: 0x60410010, Request Frame: 41460, Response Frame: 41614, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:02, Data: 0x60430010, Request Frame: 41639, Response Frame: 41786, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:03, Data: 0x30221110, Request Frame: 41791, Response Frame: 41958, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:04, Data: 0x30210c10, Request Frame: 41963, Response Frame: 42130, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:05, Data: 0x30210210, Request Frame: 42135, Response Frame: 42302, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A00:00, Data: 0x05, Request Frame: 42307, Response Frame: 42316, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A01:00, Data: 0x00, Request Frame: 42321, Response Frame: 42330, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A02:00, Data: 0x00, Request Frame: 42335, Response Frame: 42344, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1A03:00, Data: 0x00, Request Frame: 42349, Response Frame: 42376, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1600:00, Data: 0x00, Request Frame: 42381, Response Frame: 42496, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1600:01, Data: 0x60420010, Request Frame: 42501, Response Frame: 42676, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1600:00, Data: 0x01, Request Frame: 42681, Response Frame: 42690, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1601:00, Data: 0x00, Request Frame: 42695, Response Frame: 42704, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1601:01, Data: 0x60400010, Request Frame: 42709, Response Frame: 42842, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1601:02, Data: 0x60600008, Request Frame: 42847, Response Frame: 43014, Success: True, Abort Code: N/A
Configured Slave Address: 0x0001, Object: 0x1601:03, Data: 0x60720010, Request Frame: 43019, Response Frame: 43193, Success: True, Abort Code: N/A
```

SDO / PDO Analyzer 實作設計

01 設計流程
Reconstruction Pipeline

02 程式碼關係
Class Relationship

SDO Transaction

PDO Mapping

PDO Assignment

SECTION / 03

03 SyncManager 與 Local RAM 資料交換

解析 SyncManager Configuration，確認 PDO Process Data 實際使用的 ESC Local RAM Buffer。

SyncManager 功能與 PDO Assignment 關係

01
SyncManager
Local RAM Control

02
RxPDO Path
0x1C12 → SM2

03
TxPDO Path
0x1C13 → SM3

PDO Mapping

↓

PDO Assignment

↓

SyncManager

↓

ESC Local RAM

0x1C12 / 0x1C13 決定目前使用哪些 PDO Mapping；
SyncManager 則管理 Process Data 在 ESC Local RAM 中的實際資料區域與存取方式。

SyncManager Analyzer 實作設計

01 設計流程
Reconstruction Pipeline

02 程式碼關係
Class Relationship

SyncManager Configuration Reconstruction

SM Register → SyncManager Configuration → Local RAM Layout

SyncManager Analyzer 重建成果

```
[PDO Assignment Summary]

Station Address: 0x0001
Assignment Object: 0x1C12
Type: RxPDO Assignment
Active Entries: 1

Entry  Mapping Object
-----  --------------
01     0x1600

Station Address: 0x0001
Assignment Object: 0x1C13
Type: TxPDO Assignment
Active Entries: 1

Entry  Mapping Object
-----  --------------
01     0x1A00

[SyncManager Summary]

Station Address: 0x0001
SyncManager: SM2
```

SECTION / 04

04 FMMU 與 Logical Address Mapping

解析 FMMU Configuration，建立 EtherCAT Logical Address 與 ESC Physical RAM 的映射關係。

FMMU Analyzer 實作設計

01
FMMU
Logical Mapping

02
設計流程
Reconstruction Pipeline

03
程式碼關係
Class Relationship

04
重建成果
Analyzer Result

FMMU（Fieldbus Memory Management Unit）是 ESC 內的硬體單元，負責把 EtherCAT Logical Address 空間映射到 ESC 的 Physical / Local RAM 區域。Master 因此能以連續邏輯位址讀寫 Process Data，不必直接處理各 Slave 內部 RAM 的實體位址。

PDO Assignment 決定使用哪些 PDO Mapping；SyncManager 管理這批 Process Data 實際使用的 Local RAM Buffer；FMMU 再把該 RAM 區域映射到 EtherCAT Logical Address。

`PDO Mapping / Assignment → SyncManager → FMMU → Process Data`

Field	Meaning
`Logical Start Address`	EtherCAT logical address space 的映射起點
`Length`	此 FMMU mapping 涵蓋的資料長度
`Physical Start Address`	對應 ESC Local RAM 的起始位址
`Direction / Orientation`	定義 logical read / write 與 local access 的方向
`Enabled / Activate`	啟用或停用此 FMMU mapping

SyncManager 管理實體 Local RAM，FMMU 管理 Logical Address 到 Local RAM 的映射。兩者串接後，Master 才能透過 LRW、LRD、LWR 等 Process Data 命令操作 PDO 資料。

EtherCAT 圖解分析

FIGURE / 04

[Diagram Canvas]

Class Diagram / Architecture Diagram / Data Path Figure

REPLACE WITH IMG / SVG / FIGURE / HTML DIAGRAM

FOCUS POINTS

01 **Diagram Purpose**
Explain the main relationship between modules.

02 **Key Components**
Highlight the major classes or architecture blocks.

03 **Reading Order**
Describe how to read the diagram efficiently.

04 **Implementation Relevance**
Connect the diagram to the current development work.

EtherCAT 資料路徑概覽

[FIGURE]

KNOWLEDGE POINTS

01
FMMU
Logical Address
→ ESC local

02
SyncManager
Consistent data
exchange control

03
Process RAM
Stores mapped
process data

04
DATA PATH
Logical Address → FMMU →
SM → RAM → Application